Memo: Six-Group Results Analysis

TO: Dean **FROM:** Bright Joses **DATE:** 14 November 2026 **RE:** Six-group results analysis, COSC 423 results file

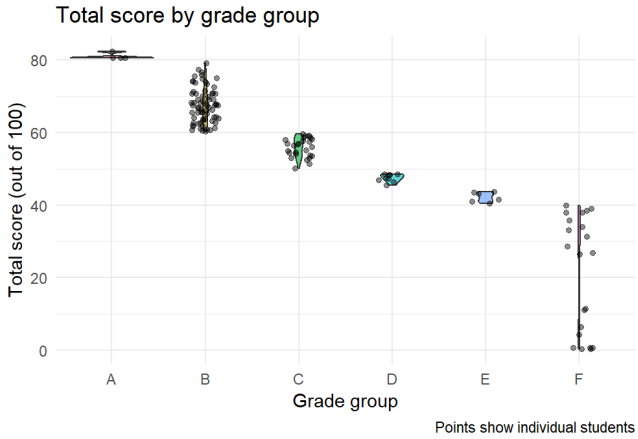
Executive summary

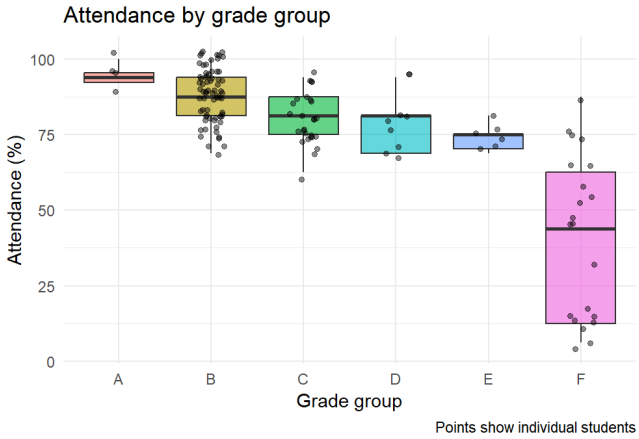
We analysed 136 students, who fall into six groups by final letter grade (A to F). Total score differs strongly across groups ($F = 148.2$, $p < 0.001$), but because grades are assigned from the score, this result is built in and says little by itself. The more useful question is whether **attendance**, which does not determine grade, separates the groups. It does, but only for one group: students graded F attended far less than everyone else, while groups A to E look alike.

Key findings

- **Attendance:** overall difference across groups is significant ($F = 42.7$, $p < 0.001$). After Tukey correction, 5 of 15 pairwise comparisons are significant, and **all five involve Group F**. Group F averaged 41.1% attendance against 87.9% for Group B (difference -46.8 points, 95% CI [-56.2, -37.4], $p < 0.001$). No other pair differs after correction.
- **Scores:** 14 of 15 pairs differ after Tukey correction; only D vs E is not significant. Group means fall steadily from 81 (A) to 21.2 (F).
- **Group F is the problem group:** 21 students (15%) averaged 21.2, with a very wide spread (SD 16), compared with SDs of 1 to 5 in the other groups.
- **Missing data:** 0 of 6392 cells (0%) are missing, so there is nothing to impute. Some students recorded zeros (did not attend or sit assessments); these are real values, not missing data.

Group	n	Score mean	Score SD	Score median	Attend. mean %	Attend. SD %
A	4	81.0	0.9	80.6	93.8	5.1
B	69	67.6	4.7	67.5	87.9	8.8
C	27	55.9	2.6	56.4	80.3	8.4
D	9	47.5	1.1	47.5	79.2	9.9
E	6	42.2	1.4	42.4	74.0	4.7
F	21	21.2	16.0	26.9	41.1	26.7





Implications

1. Attendance is the clearest early warning sign. Low attendance is concentrated in Group F, so monitoring attendance from the first weeks could identify at-risk students before the final exam.
2. Support is most useful for the students with near-zero attendance. Groups A to E do not differ meaningfully in attendance, so attendance alone will not separate a B student from a C student.
3. The scores of Groups A to E are tight within group, while Group F is not; a few F students are very low (near zero), and they should be followed up individually.

Methods and checks

One-way ANOVA with Tukey HSD for all 15 pairwise comparisons (family-wise alpha = 0.05). Because group variances are unequal (Levene-style test $p < 0.001$) and the data are not normal, I confirmed the results with Welch ANOVA (score $p < 0.001$; attendance $p < 0.001$) and Kruskal-Wallis (score $p < 0.001$; attendance $p < 0.001$); all agree. Outliers (1.5 x IQR: 9 in total score, 11 in attendance) were **kept**, since they are real students rather than entry errors; excluding the score outliers leaves the ANOVA conclusion unchanged ($F = 213.2$, still 14 of 15 pairs significant). Two CGPA ranges in the file are malformed (CSA112, CSA135) and were not used in the analysis.

Limitations

- **Circularity:** grade is derived from total score, so score differences between grade groups are expected by construction.
- **Small groups:** Group A has 4 students and Group E has 6, which limits precision (wide intervals for any comparison involving A).
- **Observational data:** one cohort, one course; associations with attendance do not show that attendance causes better results.
- **Grouping assumption:** the file has no separate group column; I used letter grade as the six groups. If a different grouping was intended, the script needs only one line changed.

Code and data: `lab2_analysis.R`, `COSC_423_Results.csv`; figures `figure_score_by_group.png` and `figure_attendance_by_group.png`; table `group_summary.csv`.